

Differential Diagnosis Copilot — Prototype Eval

MTSamples 37-case curated benchmark · Claude Haiku 4.5 / Sonnet 4.6 / Opus 4.7 · 2026-04-27

Headline. On a human-curated, leak-free set of 37 ambulatory diagnostic encounters, Claude Sonnet 4.6 ranked the correct diagnosis first in **84%** of cases and inside the top 5 in **95%**, at a total run cost of **\$0.38**. Claude Opus 4.7 cost 6.6× more and did not improve accuracy. Claude Haiku 4.5 reached 76% hit@1 at **\$0.12** — strong enough for development iteration. The bigger model is not the bottleneck; data curation and prompting are.

Results

Metric	Haiku 4.5	Sonnet 4.6	Opus 4.7
hit@1 (correct dx ranked first)	28 / 37 (76%)	31 / 37 (84%)	30 / 37 (81%)
hit@3 (correct dx in top 3)	34 / 37 (92%)	35 / 37 (95%)	35 / 37 (95%)
hit@5 (correct dx in top 5)	36 / 37 (97%)	35 / 37 (95%)	36 / 37 (97%)
Complete misses	1	2	1
Input tokens (total)	47,392	47,392	67,838
Output tokens (total)	14,076	15,881	20,084
Run cost (USD)	\$0.12	\$0.38	\$2.52
Cost per hit@1	\$0.004	\$0.012	\$0.084

Pricing reflects standard Anthropic API rates (no batch discount, no prompt caching). Both savings would compound the Sonnet cost advantage.

Methodology

Source. Public MTSamples corpus of de-identified clinical transcription samples. 199 cases fetched across 17 specialties, automatically filtered to remove procedure notes, follow-ups with pre-stated diagnoses, pediatric cases, trauma, and counseling visits.

Sanitization. Each chart was stripped of its Assessment / Impression / Plan section, the Sample-Name and Description preamble (which leak the diagnosis), and footer scaffolding. Charts without a real HPI / Chief Complaint / Reason-for-Consult section were dropped.

Human curation. 22 exact-content duplicates removed. The remaining 125 unique cases were read individually and labeled GOOD (37), ACCEPTABLE (35), or BAD (53). Only the 37 GOOD cases — which test a real differential rather than recall a stated diagnosis — were used in this run.

Eval loop. For each chart the model returned a ranked top-5 differential with one-line reasoning per item. A separate Haiku call judged whether each candidate matched any of the case's gold terms (extracted from the MTSamples keywords list plus the title-derived primary complaint). Lowest matching rank was recorded.

What this proves — and what it doesn't

Proves. The end-to-end loop works: chart in, ranked DDx out, automated scoring against ground truth. The frontier models reason capably about ambulatory presentations on this curated set, and Sonnet outperforms Opus at materially lower cost on this workload. Cost per case (~\$0.01 with Sonnet) is well within budget for a real pilot.

Does not prove. n=37 is small — confidence intervals are wide. MTSamples is transcription-clean and English-stylized; real EHR notes are messier, longer, and contain copy-forwarded artifacts. Cases were curated for fairness; production charts will include follow-ups, multi-issue visits, and ambiguous presentations. The LLM-judge introduces synonym-matching noise; some near-misses may be true matches and vice-versa.

Next. Build the synthetic-EHR pipeline (Synthea + LLM-generated notes) for longitudinal stress-testing, run a hard-mode benchmark against NEJM Case Records, and seed the first design-partner pilot to evaluate on real de-identified ambulatory data.

Reproducibility — all code, prompts, ground-truth labels, and per-case results are in the `ml/prototype_a/` directory of this repository. Raw MTSamples downloads are gitignored but regenerable with `python fetch_cases.py` & `python prepare_cases.py`.

Worked examples — what the LLM read & what it answered

Two cases from the 37-case set: a routine ambulatory presentation and a complex multi-system reasoning challenge. For each, the full sanitized chart shown to the model appears verbatim in the gray panel; Haiku 4.5's ranked top-5 differential — exactly as returned, including its one-line reasoning per item — is shown below.

Case 01 · Abdominal pain consult — clean hit at rank 1

Correct diagnosis (gold): diverticulitis (primary), sigmoid colon, lower quadrant pain

INPUT — sanitized chart shown to the model:

CHIEF COMPLAINT:

Abdominal pain.

HISTORY OF PRESENT ILLNESS:

The patient is a 71-year-old female patient of Dr. X. The patient presented to the emergency room last evening with approximately 7- to 8-day history of abdominal pain which has been persistent. She was seen 3 to 4 days ago at ABC ER and underwent evaluation and discharged and had a CT scan at that time and she was told it was "normal." She was given oral antibiotics of Cipro and Flagyl. She has had no nausea and vomiting, but has had persistent associated anorexia. She is passing flatus, but had some obstipation symptoms with the last bowel movement two days ago. She denies any bright red blood per rectum and no history of recent melena. Her last colonoscopy was approximately 5 years ago with Dr. Y. She has had no definite fevers or chills and no history of jaundice. The patient denies any significant recent weight loss.

PAST MEDICAL HISTORY:

Significant for history of atrial fibrillation, under good control and now in normal sinus rhythm and on metoprolol and also on Premarin hormone replacement.

PAST SURGICAL HISTORY:

Significant for cholecystectomy, appendectomy, and hysterectomy. She has a long history of known grade 4 bladder prolapse and she has been seen in the past by Dr. Chip Winkel, I believe that he has not been re-consulted.

ALLERGIES:

SHE IS ALLERGIC OR SENSITIVE TO MACRODANTIN.

SOCIAL HISTORY:

She does not drink or smoke.

REVIEW OF SYSTEMS:

Otherwise negative for any recent febrile illnesses, chest pains or shortness of breath.

PHYSICAL EXAMINATION:

GENERAL: The patient is an elderly thin white female, very pleasant, in no acute distress.

VITAL SIGNS: Her temperature is 98.8 and vital signs are all stable, within normal limits.

HEENT: Head is grossly atraumatic and normocephalic. Sclerae are anicteric. The conjunctivae are non-injected.

NECK: Supple.

CHEST: Clear.

HEART: Regular rate and rhythm.

ABDOMEN: Generally nondistended and soft. She is focally tender in the left lower quadrant to deep palpation with a palpable fullness or mass and focally tender, but no rebound tenderness. There is no CVA or flank tenderness, although some very minimal left flank tenderness.

PELVIC: Currently deferred, but has history of grade 4 urinary bladder prolapse.

EXTREMITIES: Grossly and neurovascularly intact.

LABORATORY VALUES:

White blood cell count is 5.3, hemoglobin 12.8, and platelet count normal. Alkaline phosphatase elevated at 184. Liver function tests otherwise normal. Electrolytes normal. Glucose 134, BUN 4, and creatinine 0.7.

DIAGNOSTIC STUDIES:

EKG shows normal sinus rhythm.

[ASSESSMENT REMOVED]

OUTPUT — Haiku 4.5's ranked differential (hit at rank 1):

- 1. Diverticulitis** — 7-8 day history of persistent left lower quadrant abdominal pain with focal tenderness and palpable mass/fullness, plus recent obstipation and anorexia in elderly patient despite prior normal CT.
- 2. Colorectal malignancy** — Persistent 7-8 day abdominal pain with left lower quadrant mass on palpation, anorexia, and obstipation symptoms in 71-year-old with last colonoscopy 5 years prior.
- 3. Sigmoid volvulus** — Elderly thin female with obstipation, persistent abdominal pain, left lower quadrant fullness on exam, and history of factors predisposing to volvulus including pelvic pathology (grade 4 bladder prolapse).
- 4. Inflammatory bowel disease (Crohn's disease or ulcerative colitis)** — Persistent 7-8 day abdominal pain with left lower quadrant tenderness and mass, anorexia, and obstipation, though lack of bloody diarrhea makes this less likely.
- 5. Ureterolithiasis or pyelonephritis** — Left-sided abdominal pain with minimal left flank tenderness and history of bladder prolapse, though normal urinalysis findings and stable vital signs make infection less likely.

Case 135 · Multi-system presentation — clean hit at rank 1

Correct diagnosis (gold): vasculitis / granulomatosis with polyangiitis, hemoptysis, polyarthralgia

INPUT — sanitized chart shown to the model:

Chief Complaint:

coughing up blood and severe joint pain.

History of Present Illness:

The patient is a 37 year old African American woman with history of chronic allergic rhinitis who presents to an outpatient clinic with severe pain in multiple joints and hemoptysis for 1 day. The patient was at her baseline state of health until 2 months prior to admission when her usual symptoms of allergic rhinitis worsened. In addition to increased nasal congestion and drainage, she also began having generalized fatigue, malaise, and migratory arthralgias involving bilateral wrists, shoulders, elbows, knees, ankles, and finger joints. She also had intermittent episodes of swollen fingers that prevented her from making a fist. Patient denied recent flu-like illness, fever, chills, myalgias, or night sweats. Four weeks after the onset of arthralgias patient developed severe bilateral eye dryness and redness without any discharge. She was evaluated by an ophthalmologist and diagnosed with conjunctivitis. She was given eye drops that did not relieve her eye symptoms. Two weeks prior to admission patient noted the onset of rust colored urine. No bright red blood or clots in the urine. She denied having dysuria, decreased urine output, abdominal pain, flank pain, or nausea/vomiting. Patient went to a community ER, and had a CT Scan of the abdomen that was negative for kidney stones. She was discharged from the ER with Bactrim for possible UTI. During the next week patient had progressively worsening arthralgias to the point where she could hardly walk. On the day of admission, she developed a cough productive of bright red blood associated with shortness of breath and nausea, but no chest pain or dizziness. This prompted the patient to go see her primary care physician. After being seen in clinic, she was transferred to St. Luke's Episcopal Hospital for further evaluation.

Past Medical History:

Allergic rhinitis, which she has had for many years and treated with numerous medications. No history of diabetes, hypertension, or renal disease. No history tuberculosis, asthma, or upper airway disease.

Past Surgical History:

Appendectomy at age 21. C-Section 8 years ago.

Ob/Gyn: G2P2; last menstrual period 3 weeks ago. Heavy menses due to fibroids.

Social History:

Patient is married and lives with her husband and 2 children. Works in a business office. Denies any tobacco, alcohol, or illicit drug use of any kind. No history of sexually transmitted diseases. Denies exposures to asbestos, chemicals, or industrial gases. No recent travel. No recent sick contacts.

Family History:

Mother and 2 maternal aunts with asthma. No history of renal or rheumatologic diseases.

Medications:

Allegra 180mg po qd, Zyrtec 10mg po qd, Claritin 10mg po qd

No herbal medication use.

Allergies:

No known drug allergies.

Review of systems:

No rashes, headache, photophobia, diplopia, or oral ulcers. No palpitations, orthopnea or PND. No diarrhea, constipation, melena, bright red blood per rectum, or pale stool. No jaundice. Decreased appetite, but no weight loss.

Physical Examination:

VS: T 100.2F BP 132/85 P 111 RR 20 O2 Sat 95% on room air

GEN: Well-developed woman in no apparent distress.

SKIN: No rashes, nodules, ecchymoses, or petechiae.

LYMPH NODES: No cervical, axillary, or inguinal lymphadenopathy.

HEENT: Pupils equally round and reactive to light. Extra-ocular movements intact. Anicteric sclerae. Erythematous sclerae and pale conjunctivae. Dry mucous membranes. No oropharyngeal lesions. Bilateral tympanic membranes clear. No nasal deformities.

NECK: Supple. No increased jugular venous pressure. No thyromegaly.

CHEST: Decreased breath sounds throughout bilateral lung fields with occasional diffuse crackles. No wheezes or rales.

CV: Tachycardic. Regular rhythm. No murmurs, gallops, or rubs.

ABDOMEN: Soft with normal active bowel sounds. Non-distended and non-tender. No masses palpated. No hepatosplenomegaly.

RECTAL: Brown stool. Guaiac negative.

EXT: No clubbing, cyanosis, or edema. 2+ pulses bilaterally. Tenderness and mild swelling of bilateral wrists, MCPs and PIPs with decreased range of motion and grip function. Bilateral wrists warm without erythema. Bilateral elbows, knees, and ankles tender to palpation with decreased range of motion, but no erythema, warmth, or swelling of these joints.

NEURO: Cranial nerves intact. 2+ DTRs bilaterally and symmetrically. Motor strength and sensation are within normal limits.

STUDIES:

Chest X-ray (10/03):

Suboptimal inspiratory effort. No evidence of pneumonic consolidation, pleural effusion, pneumothorax, or pulmonary edema. Cardiomeastinal silhouette is unremarkable.

CT Scan of Chest (10/03):

Prominence of the bronchovascular markings bilaterally with a nodular configuration. There are mixed ground glass interstitial pulmonary infiltrates throughout both lungs with a perihilar predominance. Aortic arch is of normal caliber. The pulmonary arteries are of normal caliber. There is right paratracheal lymphadenopathy. There is probable bilateral hilar lymphadenopathy. Trachea and main stem bronchi are normal. The heart is of normal size.

Renal Biopsy:

Microscopic Description : Ten glomeruli are present. There are crescents in eight of the glomeruli. Some of the glomeruli show focal areas of apparent necrosis with fibrin formation. The interstitium consists of a fairly dense infiltrate of lymphocytes, plasma cells with admixed eosinophils. The tubules for the most part are unremarkable. No vasculitis is identified.

Immunofluorescence Description : There are no staining for IgG, IgA, IgM, C3, Kappa, Lambda, C1q, or albumin.

Electron Microscopic Description : Mild to moderate glomerular, tubular, and interstitial changes. Mesangium has multifocal areas with increased matrix and cells.

There is focal mesangial interpositioning with the filtration membrane. Interstitium has multifocal areas with increased collagen. There are focal areas with interstitial aggregate of fibrin. Within the collagen substrate are infiltrates of lymphocytes, plasma cells, eosinophils, and macrophages. The glomerular sections evaluated show no electron-dense deposits in the filtration membrane or mesangium.

Microscopic Diagnosis: Pauci-immune crescentic glomerulonephritis with eosinophilic interstitial infiltrate.

OUTPUT — Haiku 4.5's ranked differential (hit at rank 1):

1. Granulomatosis with polyangiitis (GPA) — Classic triad of upper respiratory involvement (worsening allergic rhinitis), pulmonary findings (hemoptysis, bilateral infiltrates with perihilar predominance and hilar lymphadenopathy on CT), and pauci-immune crescentic glomerulonephritis with eosinophilic infiltration on renal biopsy in an African American woman with polyarthralgias.

2. Eosinophilic granulomatosis with polyangiitis (EGPA) — Pauci-immune crescentic glomerulonephritis with prominent eosinophilic interstitial infiltrate on renal biopsy, pulmonary infiltrates with hemoptysis, systemic symptoms including migratory polyarthralgias, and constitutional symptoms suggest EGPA variant.

3. Microscopic polyangiitis — Pauci-immune crescentic glomerulonephritis with negative immunofluorescence, hemoptysis with bilateral pulmonary infiltrates, and systemic polyarthralgias are consistent with microscopic polyangiitis, though absence of typical rash and prominent eosinophils make this less likely.

4. Anti-glomerular basement membrane disease (anti-GBM) — Pauci-immune crescentic glomerulonephritis with hemoptysis and pulmonary-renal syndrome presentation; however, negative immunofluorescence and prominent eosinophilic infiltration make anti-GBM less likely.

5. Systemic lupus erythematosus — Polyarthralgias, hemoptysis, glomerulonephritis, and constitutional symptoms in a young African American woman are compatible with SLE, though negative immunofluorescence staining and pauci-immune pattern on renal biopsy make this less likely.